

Differentiation - Set 2 - Solutions

Specification MM6.1-MM6.3, MM8.5.

Answer key

Question	Answer	Topic
1	C	Differentiation
2	E	Differentiation
3	A	Differentiation
4	E	Differentiation
5	C	Differentiation
6	E	Differentiation
7	B	Differentiation
8	G	Differentiation
9	C	Differentiation
10	F	Differentiation

Worked solutions

Question 1 DIF-18 • Differentiation • spec MM6.3

Find the minimum value of $x + \frac{4}{x}$ for $x > 0$.

Answer: C

Derivative $1 - \frac{4}{x^2} = 0$ at $x = 2$; value $2 + 2 = 4$ (second derivative $\frac{8}{x^3} > 0$, so a minimum).

Question 2 DIF-20 • Differentiation • spec MM6.3

Find the sum of the y-coordinates of the stationary points of the curve $y = 2x^3 - 9x^2 + 12x - 3$.

Answer: E

$y' = 6x^2 - 18x + 12 = 6(x-1)(x-2)$: stationary points at $x = 1$ ($y = 2 - 9 + 12 - 3 = 2$) and $x = 2$ ($y = 16 - 36 + 24 - 3 = 1$). Sum 3.

Question 3 DIF-08 • Differentiation • spec MM6.2

Given $f(x) = x^{\frac{3}{2}} - 6x^{\frac{1}{2}}$ for $x > 0$, find the value of x at which f has its minimum.

Answer: A

$f'(x) = \frac{3}{2}x^{1/2} - 3x^{-1/2} = \frac{3}{2\sqrt{x}}(x-2) = 0$ at $x = 2$; f' changes from negative to positive there, so it is a minimum.

Question 4 DIF-11 • Differentiation • spec MM6.3

Which of the following statements about $f(x) = x^3 - 3x$ is/are true?

- I f is increasing for all $x > 1$.
- II f has a local maximum at $x = -1$.
- III $f(x) \geq -2$ for all real x .

Answer: E

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$: positive for $x > 1$ (I true); $f'' = 6x < 0$ at $x = -1$ (II true). III false: the local minimum $f(1) = -2$ is not a global minimum since $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ (e.g. $f(-3) = -18$).

Question 5 DIF-13 • Differentiation • spec MM6.1

$f(x)$ is a polynomial with $f'(x) = (x-2)^2(x+1)$. Which of the following is true?

Answer: C

f' changes sign from negative to positive at $x = -1$ (local minimum). At $x = 2$ the factor $(x-2)^2$ does not change sign, so f' stays positive: a stationary point but not a turning point (a point of inflexion).

Question 6 DIF-15 • Differentiation • spec MM6.3

The volume of a cuboid with a square base of side x is $V = 4x^2 - x^3$ for $0 < x < 4$. Find the maximum volume.

Answer: E

$V' = 8x - 3x^2 = x(8-3x) = 0 \Rightarrow x = \frac{8}{3}$. $V = \frac{64}{9} \cdot 4 - \frac{512}{27} = \frac{768-512}{27} = \frac{256}{27}$.

Question 7 DIF-16 • Differentiation • spec MM6.3

The tangent to $y = x^3$ at the point (a, a^3) , where $a \neq 0$, meets the curve again at the point with x -coordinate

Answer: B

Tangent: $y = 3a^2x - 2a^3$. Solve $x^3 - 3a^2x + 2a^3 = 0$; $x = a$ is a double root, so $(x - a)^2(x + 2a) = 0$: the other intersection is $x = -2a$.

Question 8 DIF-19 • Differentiation • spec MM6.1

Which of the following statements is/are true for every function f that has a derivative at every real number?

- I If $f'(a) = 0$ then f has a maximum or a minimum at $x = a$.
- II If f has a local minimum at $x = a$ then $f'(a) = 0$.
- III If $f'(x) > 0$ for all x , then $f(3) > f(2)$.

Answer: G

I is false: $f(x) = x^3$ has $f'(0) = 0$ but no turning point. II is true (a stationary point is required at an interior local extremum of a differentiable function). III is true: a strictly increasing function has $f(3) > f(2)$.

Question 9 X2-02 • Differentiation • spec MM6.3

Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?

Answer: C

If $f' > 0$ everywhere, f is strictly increasing and (being an odd-degree polynomial) crosses the axis exactly once: P is sufficient. But $f(x) = x^3$ has exactly one root and $f'(0) = 0$, so P is not necessary.

Question 10 X2-16 • Differentiation • spec MM6.3

The function f has $f'(x) = (x - 1)(x - 3)^2$. Which of the following statements is/are true?

- I f has a local minimum at $x = 1$.
- II f has a local maximum at $x = 3$.
- III f is increasing for all $x > 1$.

Answer: F

f' changes from negative to positive at $x = 1$ (minimum). At $x = 3$ the squared factor keeps $f' \geq 0$ on both sides, so no turning point: II is false, and $f' \geq 0$ for $x > 1$ (zero only at 3) makes III true.